

1 There are a variety of methods used to estimate the water potential of plant tissues.

(a) A student decided to estimate the water potential of potato tuber tissue using the method described.

- A number of test-tubes were set up, each containing the same volume of sucrose solution.
- Each test-tube contained a different concentration of sucrose solution.
- Pieces of potato tissue were placed in each test-tube.
- After a period of time, the potato tissue was removed from the test-tubes.

The student then measured the change in density of the sucrose solutions left in each test-tube. The more water present in a sucrose solution, the less dense it is.

- Sucrose solutions of the same concentrations as those of the **original** test-tubes were prepared.
- Each of these sucrose solutions was coloured with methylene blue.
- One drop of the coloured sucrose solution of a known concentration was released into the middle of the test-tube whose sucrose concentration was originally the same as that of the coloured sucrose drop.

The three possible outcomes that the student would see are shown in Fig. 1.1.

- 1 The drop will rise if the coloured sucrose drop is less dense than the sucrose solution in the test-tube.
- 2 The drop will remain where it is released if the coloured sucrose drop is similar in density to the sucrose solution in the test-tube.
- 3 The drop will sink if the coloured sucrose drop is more dense than the sucrose solution in the test-tube.

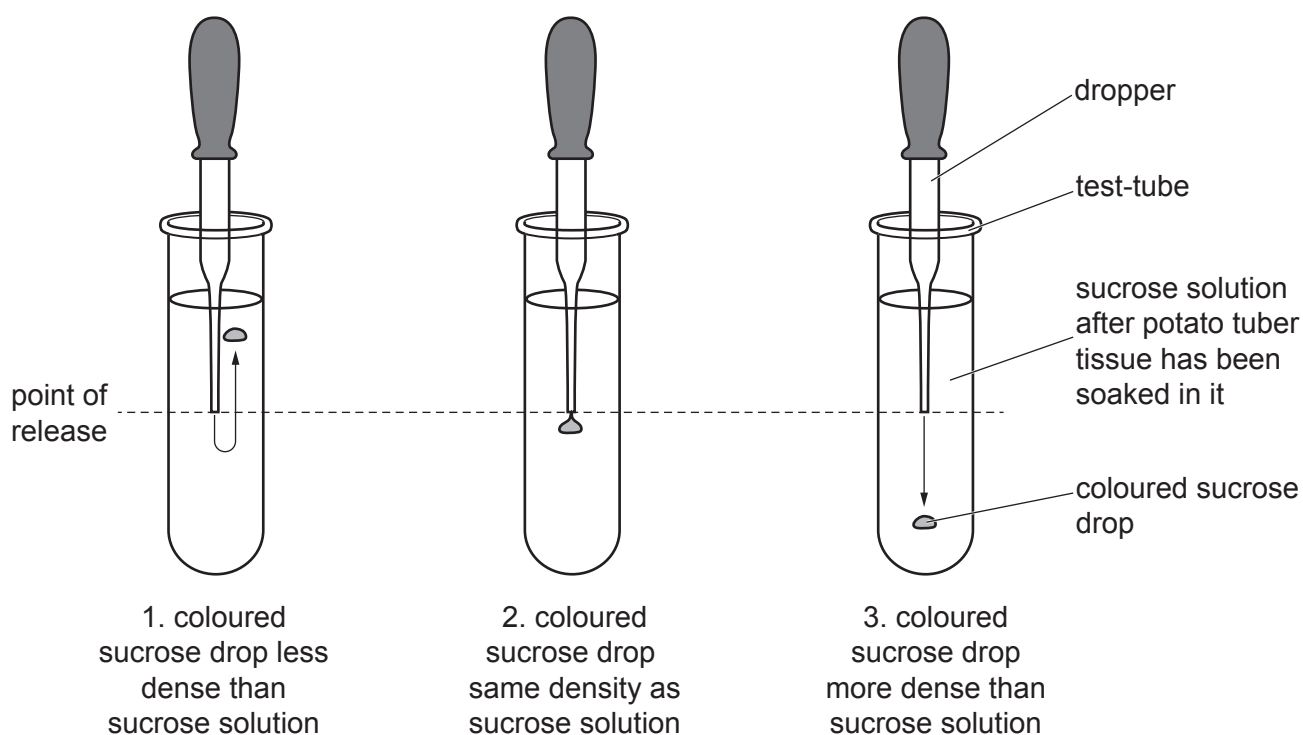


Fig. 1.1

(i) Identify the independent variable in the experiment.

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..... [1]

(ii) Predict the result if the water potential of the potato tissue was **higher** (less negative) than the surrounding sucrose solution.

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..... [1]

(iii) Explain the prediction that you gave in (a)(ii).

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..... [2]

- (b) The student then used another method to estimate the water potential of potato tissue.

The student placed pieces of potato tissue in a range of concentrations of sucrose solutions.

The equivalent sucrose concentration was identified as the concentration of the sucrose solution in which the potato tissue did not gain or lose mass.

The equivalent sucrose concentration indicates the water potential of the potato tissue.

- (i) The student was given a stock solution of sucrose with a concentration of 1.0 mol dm^{-3} from which to make a range of dilutions. The student decided to make 20 cm^3 of each concentration of sucrose solution.

Describe how the student prepared 20 cm^3 of solution with a concentration of 0.4 mol dm^{-3} .

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..... [1]

- (ii) The student carried out a risk assessment before carrying out the experiment to find the equivalent sucrose concentration.

Suggest **one** reason why the student decided the level of risk for this method was **low to medium**.

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..... [1]

- (c) The student determined a value of 0.46 mol dm^{-3} for the equivalent sucrose concentration of the potato tissue.

Use Fig. 1.2 to determine the water potential of the potato tissue.

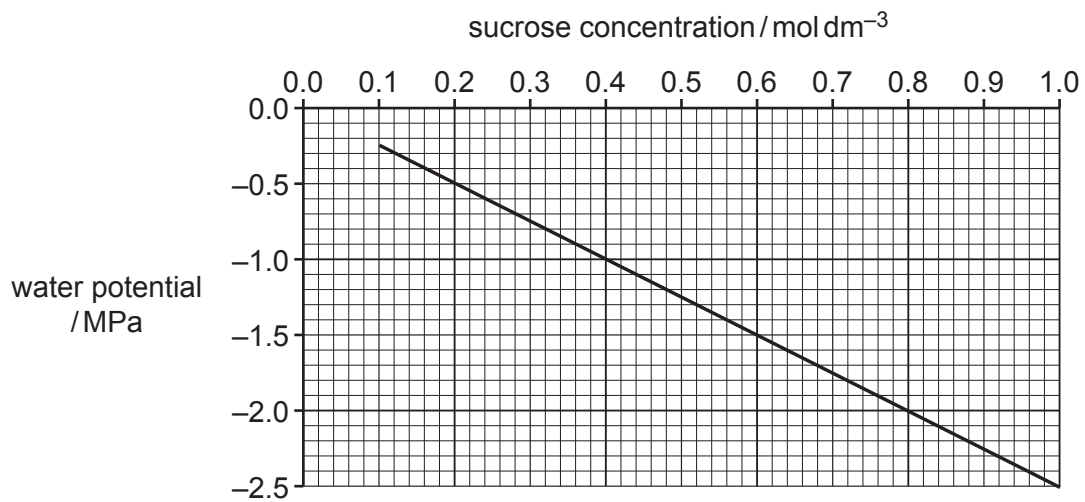


Fig. 1.2

water potential of potato tissue = MPa [1]

- (d) The student compared two types of potato, Record and Kerr's Pink, to see if there was a significant difference between the water potential of the two potato tissues.

The student decided to carry out a *t*-test.

- (i) State **one** condition of the data that allows the student to carry out a *t*-test.

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 [1]

- (ii) State the null hypothesis for the *t*-test that the student carried out.

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 [1]

(iii) The student tested 20 samples for each type of potato.

Calculate the number of degrees of freedom (ν) for the t -test using the formula given.

$$\nu = n_1 + n_2 - 2$$

Key to symbols:

n = sample size (number of observations)

number of degrees of freedom = [1]

Table 1.1 shows the critical values for t at 0.05 probability.

Table 1.1

degrees of freedom	36	37	38	39	40	41	42	43	44
critical value at 0.05 probability	2.028	2.026	2.024	2.023	2.021	2.020	2.018	2.017	2.015

(iv) The student calculated t as 1.890.

With reference to Table 1.1 and the calculated value of t , make a conclusion about the water potentials of the two types of potato tissue, Record and Kerr's Pink.

Explain your reasoning.

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..... [3]

[Total: 19]